

Runbook — Start here

This is the front page of the runbook for the off-grid solar system. You do not need to understand how the system works to use it. Find the line that matches what you are seeing, go to that page, and follow the numbered steps. Each page tells you when to stop and call for help.

▲ STOP — three rules that apply to every page

- **Never open an inverter, a battery pack, or the generator's control panel door while anything is running.** The only things you touch are breakers, the round PV switch on the side of each inverter, the power button, the touchscreen, and the generator's buttons.
- **The solar panels and their wires are live whenever there is light on them — even moonlight.** Turning a breaker off does not make the wires before it safe.
- **Lights are named by their printed label** (ALM, RUN, DC, AC, Normal, Alarm) and by whether they are steady or flashing — you never need to tell colors apart to use these pages.
- **If you are not sure, stop and call.** Nothing in this system gets worse by waiting an hour for a phone call. Some things get much worse if the wrong button is pressed.

What you see → which page

What you see	Go to
House has no power; inverter screens dark, or the indicator labelled Alarm is lit	Batteries page, Procedure 3 — then Inverter failed page if only one inverter is dark
Inverter screen shows Low Batt (the battery symbol changes from its normal look)	Batteries page, Procedure 2
Inverter has switched the house power off to protect the batteries (screen shows the battery symbol changed, or is dark)	Batteries page, Procedure 3
On a battery pack, the light marked ALM is on steadily and nothing else is lit — or no lights at all	Batteries page, Procedure 4
Generator should have started but didn't	Generator page, Procedure 2
You want to run the generator now to charge	Generator page, Procedure 1
Generator is running but keeps stopping, or the inverter won't take its power	Generator page, Procedure 3
The big generator is broken and you have a portable one	Generator page, Procedure 5
One inverter's screen is dark or its Alarm indicator is lit; the other shows Normal	Inverter failed page, Procedure 1
Both inverter screens show F41 or F29	Inverter failed page, Procedure 1
SolarAssistant shows one inverter column frozen	Inverter failed page, Procedure 1
Solar production is low, or one MPPT shows 0 V	Solar panels page, Procedure 4
Snow or ice on the panels	Solar panels page, Procedure 2
A panel is cracked or damaged	Solar panels page, Procedure 5 (call first)
It is spring or autumn and the panels need tilting	Solar panels page, Procedure 3; Calendar page
Routine upkeep — what to do this month	Calendar page

The pages

Page	File
Batteries — low battery, power went out, a pack shut itself off	cheatsheet-low-battery
Generator — starting it, charging, when it won't run	cheatsheet-generator
Inverter failed — one inverter down, house still has some power	cheatsheet-inverter-failure
Solar panels — cleaning, snow, tilt, replacing a panel	cheatsheet-array-maintenance
Maintenance calendar — each month, and what to keep on hand	cheatsheet-maintenance-calendar
Technician appendix — reference tables, specs and technician-only procedures (not needed by end users)	cheatsheet-99-technician-appendix

Who to call

Who	When	Contact
Installer — Ernie Williams, Mainstream Green Solutions, Lexington TN	First call for anything on these pages	(731) 697-1665 · Ernie.williams@mainstreamgreensolutions.com · www.mainstreamgreensolutions.com
Sol-Ark Technical Support	Inverter fault codes, settings	support@sol-ark.com — 7 days a week, not 24 h
Pytes (battery maker)	A pack that will not wake after charging	ess_support@pytesgroup.com
Generator dealer	Engine faults that will not clear	Number on the generator nameplate

Where things are

Thing	Where
Inverter touchscreens and power buttons	On the front of each Sol-Ark inverter
Round PV Disconnect switch	On the side of each inverter
Battery pack lights and buttons	Front of each Pytes pack in the battery rack
Solar DC wall disconnect	Exterior wall (single-line drawing is engraved beside it)
Combiner boxes (solar breakers)	At the end of each array
Generator control panel (DSE)	Inside the generator enclosure, left rear door
Generator 120 V inlet (sump heater + battery charger)	Power-panel door on the generator
Portable generator inlet	50 A outdoor receptacle (feeds the inverters' GEN inputs)
SolarAssistant	https://pnc-home.us.solar-assistant.io